

Modernizing Legacy Applications: Maximizing the Investment

June 2007

— Underwritten, in Part, by —



Executive Summary

IT is under intense pressure to provide new ways for the business to access data and functionality running on legacy platforms. Legacy code often embodies the core processes of the company, and simply rewriting it is both prohibitively expensive and disruptive. In many cases, the best thing to do both from a performance and cost standpoint is to leave the legacy code alone.

Organizations that are SOA-enabling their legacy applications *on the legacy platform* are outperforming those that are using any other approach. They report better productivity, higher agility, and lower costs for legacy integration projects.

"We are going from 12,000 applications to hundreds of services using an SOA. The lifecycle cost savings are already flowing through in lower maintenance costs and faster development cycles. This is turning into a major win for the business, not just IT."

~ VP, Financial Services

Best in Class Performance

Four key performance indicators (KPIs) were used to identify those companies that are succeeding in their legacy modernization efforts, Aberdeen's Best In Class (BIC):

- Overall cost of legacy integration projects
- Number of new applications delivered that are based on legacy applications
- Time-to-value for customer-facing projects
- Agility of IT in responding to new business requirements

Competitive Maturity Assessment

Each organization tailors its approach to legacy modernization based on their application portfolio. The Best In Class in this study understand that modernization projects are not just about technology; they also require changes to key processes.

The typical Best In Class organization:

- Uses automation to extract business logic from legacy applications (61%)
- Is SOA-enabling their legacy applications on the legacy platform (33%)
- Is using SOA to increase the agility of the IT department in responding to new business requirements (40%)
- Has planned for the unique challenges of testing legacy application integration by putting new test processes into place (61%)
- Has SLAs in place for legacy interfaces (63%)
- Sees their legacy modernization projects as a long-term solution (91%)

Required Actions

Legacy modernization projects touch core components of your business. Take these steps to minimize the risk and maximize the investment:

- **Automate.** Most legacy applications are a mix of data and logic, and the interfaces are often not clearly defined. Using automated tools to extract business logic and data structure will let you focus on solving the business problem.
- **Plan for test.** Best In Class organizations are twice as likely as Industry Average or Laggards to have well-defined test processes for integration testing.
- **Build a services layer on the mainframe.** Exposing legacy packages through a services layer minimizes disruption to the business and reduces risk.

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Chapter One: Benchmarking the Best in Class

Aberdeen Analysis

Legacy applications often contain the organization's core business functionality and represent large capital investments in both dollars and manpower. Wholesale replacement of these applications is impractical for time and cost reasons.

Faced with competing pressures to preserve the investment in legacy applications and platforms, to provide new IT services to support changing business requirements, and to minimize disruption to existing processes, IT typically responds in one of two general ways:

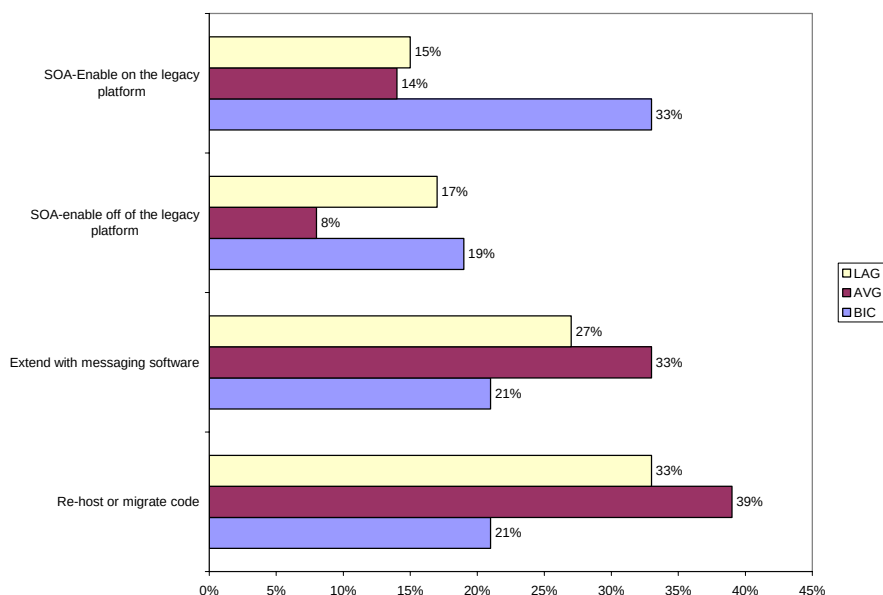
- **Extend/surround**, to provide new interfaces to the legacy application, either on the mainframe or off, in order to open it up to web services or SOA middleware;
- **Re-host**, moving the entire legacy application to a new platform while preserving the existing code base.

Figure I shows that Best In Class organizations more often choose to leave packages on the legacy platform and use software adapters or a services layer to provide access to other SOA applications.

Fast Facts

- ✓ **61%** of Best In Class organizations report a decrease in the cost of legacy platform maintenance costs
- ✓ **70%** of Best In Class companies have clearly documented legacy packages and data structures
- ✓ **90%** of Best In Class see their legacy modernization projects as a long-term, strategic plan

Figure I: What Approaches Are Being Used?

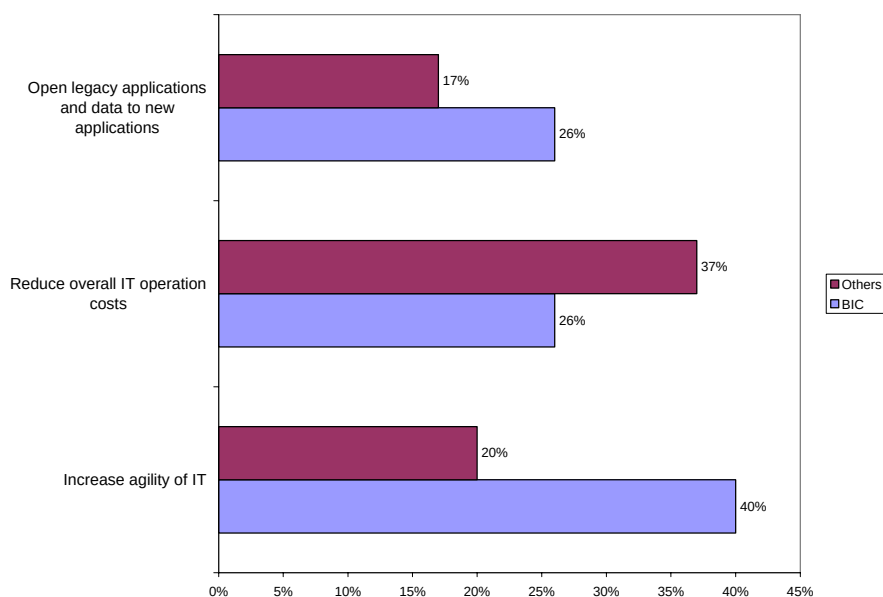


Source: Aberdeen Group, June 2007

Why Enterprises Are Focusing on Legacy Modernization

Agility is at the top of the list for Best In Class organizations trying to bring their legacy applications into a modern architecture. This means the ability to rapidly build services, based on legacy functionality and data, that provide solutions to new business problems. While cost is a consideration, it is not the driving factor for this group (Figure 2).

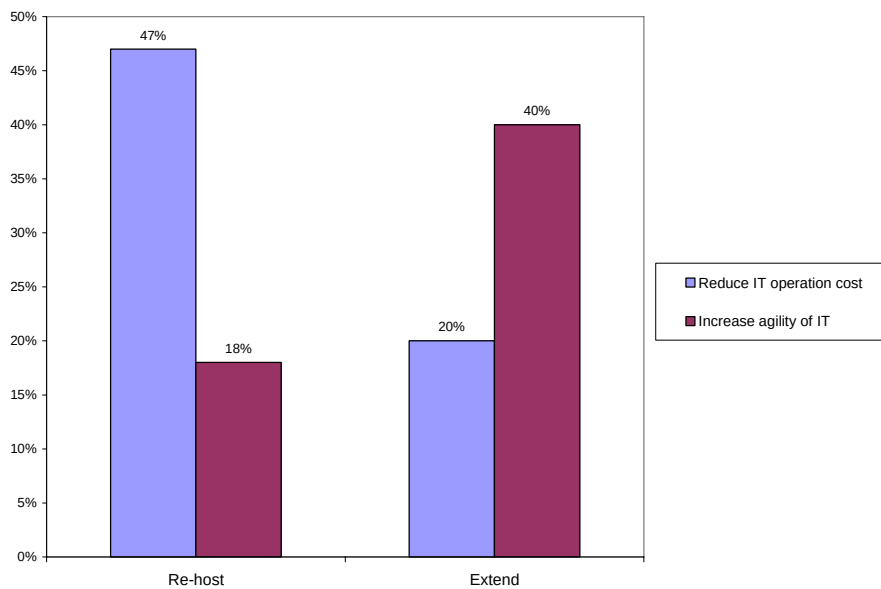
Figure 2: Top drivers for legacy modernization efforts



Source: Aberdeen Group, June 2007

Key drivers for the Industry Average and Laggard groups center more around cost, and this is reflected in their overall approach to legacy modernization. While Best In Class organizations are leveraging their SOA infrastructure to provide interfaces into their legacy applications, Industry Average and Laggard companies opt for re-hosting, which moves the legacy code base to a less expensive hardware platform (Figure 3).

Figure 3: Modernization approach reflects organizational drivers



Source: Aberdeen Group, June 2007

Operational costs can be a significant driver. As Larry Munini, President of Genesys, a human-capital management (HCM) software vendor in business for 26 years puts it, “This code is our business. It’s what we do. But at the end of the day, our customers don’t care what lights up their web browser.” Moving COBOL packages from the mainframe to less expensive Intel platforms resulted in major cost savings. Where performance was essential, VSAM files were left in place.

Choosing re-hosting on a cost basis alone may lead to disappointment. Of those organizations that chose this approach, only 19% reported a reduction in the cost of legacy integration projects. Forty two percent said that their legacy infrastructure capital costs had decreased.

Maturity Class Framework

Aberdeen used four key performance criteria to distinguish Best in Class companies from Industry Average and Laggard organizations:

- Overall cost of legacy integration projects
- Number of new applications delivered that are based on legacy applications
- Time-to-value for customer-facing projects
- Agility of IT in responding to new business requirements

With Best In Class identified, we then looked at how the group compared to the Industry Average and Laggard organizations (Table I).

Table I: Companies With Top Performance Earn “Best-in-Class” Status

Definition of Maturity Class	Mean Class Performance
<u>Best in Class:</u> Top 20% of aggregate performance scorers	• 46% saw reduction in legacy integration costs • 43% reported an increase in the number of new application delivered that were based on legacy packages • 61% saw a decrease in the cost of maintenance for legacy platforms
<u>Industry Average:</u> Middle 50% of aggregate performance scorers	• 9% saw reduction in legacy integration costs • 21% reported an increase in the number of new application delivered that were based on legacy packages • 30% saw a decrease in the cost of maintenance for legacy platforms
<u>Laggard:</u> Bottom 30% of aggregate performance scorers	• 13% saw <i>increases</i> in legacy integration costs • 8% reported a <i>decrease</i> in the number of new application delivered that were based on legacy packages • 5% saw a decrease in the cost of maintenance for legacy platforms

Source: Aberdeen Group, 2007

Best in Class PACE Model

Aberdeen uses the PACE model to profile the top performing 20% of companies – the Best In Class – in four areas:

- **Pressures:** External or internal pressures affecting an organization. These might be economic, technical, strategic, or competitive.
- **Actions:** The ways that an organization responds to pressures.
- **Capabilities:** The internal capabilities that an organization must have to take the specified actions.
- **Enablers:** Technical products, procedures, or designs necessary to carry out actions.

The Best in Class are using SOA and web services to quickly respond to new business requirements by extending legacy packages and data out to the enterprise.

Table 2: Best-in-Class PACE Framework

Pressures	Actions	Capabilities	Enablers
• Improve the agility of the IT department when responding to new business requirements	• Extend legacy code to provide new interfaces to service-based architectures • Surround legacy applications with messaging middleware • Re-host legacy code	• Services-based infrastructure is in place • Business and IT processes are aligned • Legacy applications and interfaces are well defined	• Third-party adapters for legacy systems • SOA/web services layer on the mainframe • Automated tools for extracting legacy data and business logic • Middleware platforms for rehosting

Source: Aberdeen Group, June 2007

Aberdeen Insights – Which Approach Is Best?
Aberdeen research of over 400 companies in the past 18 months has shown that Best In Class IT organizations have embraced SOA as a way to rapidly develop and deploy solutions to new business problems. In the realm of legacy applications, this approach is paying off: Companies that are leveraging their investment in SOA are outperforming their competitors by using it to build new applications based on legacy code and data. Re-hosting remains an attractive alternative for those organizations that are sensitive to cost, but only 19% of respondents who chose re-hosting reported a decrease in the overall cost of IT integration projects.

Chapter Two: Benchmarking Requirements for Success

Best In Class IT organizations are leveraging both their investment in legacy platforms and their existing SOA infrastructure. They are focusing on extending their legacy applications *on the mainframe* by building a services layer that exposes legacy data and packages.

Case Study: Standard components make the difference at CFS

Brian Digman, CIO of the Enterprise Services Division of the State of New York Department of Taxation and Finance, faced mounting costs, an aging staff of COBOL programmers, systems that weren't much younger, tens of millions of lines of legacy code, proprietary databases, and a workflow that included multiple paper steps. There were few web interfaces, and a significant backlog of work.

The vision: Adopt an SOA approach to modernization using XML to facilitate communication, provide new ways to access existing applications and data, and eliminate as much paper processing as possible. Starting with a pilot project focused on handling exceptions in sales tax processing, the team architected an SOA solution that wraps business components with new web-based interfaces, keeping COBOL in place where it made sense, and used XML as a common language for messaging. Automating the workflow helped remove paper from the picture. XML also provided ways to build tax forms on the fly on web clients, both for internal processing and for taxpayer use, and to hook into other systems such as interactive voice response. By moving most processing to memory, performance increased dramatically. On a key inbound processing step, throughput went from 18,000 transactions per hour to over 60,000.

"The ROI was staggering," says Digman. "We removed the paper, reduced our exception backlog by 60%, and lowered the cost of labor by using standard packages instead of custom code. SOA is now helping DTF transform our highest-volume system, personal income tax, from batch to transactional processing using XML, electronic forms, and middleware for business process orchestration."

Fast Facts

- ✓ Best In Class companies are **twice as likely** as other groups to use automated tools for extracting business logic and data structures
- ✓ **43%** of Best In Class organizations use third-party services adapters for legacy applications, versus 28% of the Laggards group
- ✓ **Twice as many** Best In Class companies versus all others have deployed a SOA or web services layer on the mainframe

Competitive Assessment

Survey respondents fell into one of three categories – Laggard, Industry Average, or Best in Class — based on their characteristics in five key categories: (1) process (how the organization itself changes to accommodate new business requirements); (2) organization (large-scale changes in the IT department of technology or procedure); (3) knowledge (training of key personnel in new technologies); (4) technology (selection or appropriate tools and intelligent deployment of those tools); and (5) performance management (ability of the organization to measure the benefits of technology deployment and use the results to improve key processes further).

Table 3: Competitive Framework

	Laggards	Average	Best-in-class
Process	Legacy applications and interfaces are documented		
	54%	56%	70%
	Integration processes are defined		
	45%	41%	62%
	Testing processes for legacy integration are defined		
	40%	33%	61%
Organization	Business and IT processes are aligned		
	41%	49%	60%
Knowledge	Key developers, business analysts, and architects are trained		
	38%	35%	51%
Technology	Third-party adapters for legacy applications		
	28%	36%	43%
	SOA / Web services layer on the mainframe		
	15%	12%	32%
	Automated tools for extracting business logic and data		
	19%	20%	38%
	Integrated development environments (IDEs) aimed at legacy systems		
	25%	35%	56%
Performance	Service level agreements (SLAs) for legacy interfaces are in place		
	30%	43%	63%

Source: Aberdeen Group, June 2007

Organizational Capabilities and Technology Enablers

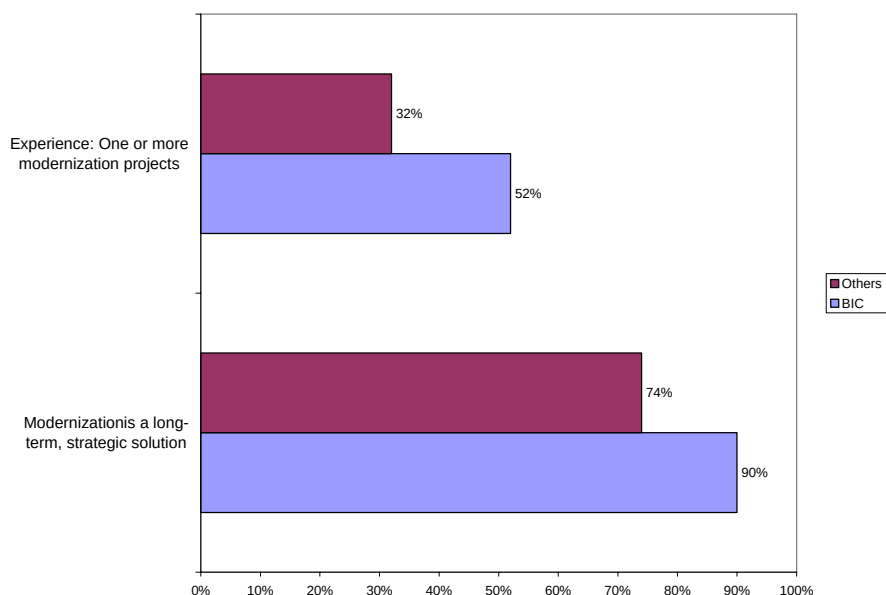
SOA has been described in many places as a journey. During the past eighteen months of research on SOA at Aberdeen, we have watched both the technologies and the organizations using them mature significantly. The Best In Class companies are still leading the charge, and with more overall experience under their belts, they are setting the pace for the rest of the industry. Over half of the Best In Class organizations are either working on or have completed at least one legacy modernization project, versus fewer than a quarter of Laggards (Figure 3).

All groups see their modernization efforts as a long-term solution. Industry Average and Laggards, trailing Best In Class in experience, are also likely to be running proof-of-concept and pilot projects.

“After looking through all of the code running on the mainframe, we discovered that 40% of it was unused...and about 10% of it nobody knew what it was supposed to do.”

~ Manager, SOA modernization vendor

Figure 4: Organizational view of modernization



Source: Aberdeen Group, June 2007

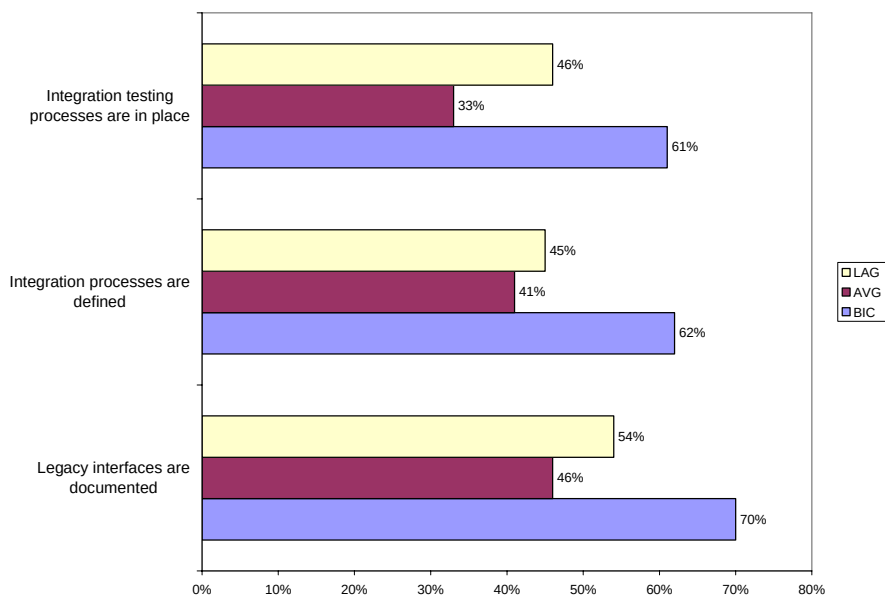
Process

The Best In Class organizations have, in general, been on the SOA journey longer and therefore have had more time to establish policies around their application work. This maturity is reflected in an emphasis on process versus the Industry Average and Laggard groups (Figure 4).

One of the challenges in any work on legacy code is that, over time, layers upon layers of application logic have been added, and it isn't always easy to tease out the functionality that is actually being used. As the manager at a modernization vendor said of a client's project, "After looking through all of the code running on the mainframe, we discovered that around 40% of it was unused...and about 10% of it nobody knew what it was supposed to do."

Testing is another stumbling block encountered by Industry Average and Laggard organizations. Trying to apply old test methodology to new architectures such as SOA and web services will quickly result in friction in the lifecycle process. Best In Class companies are about **twice as likely** as others to have clearly defined testing processes in place.

Figure 5: Best in Class place an emphasis on process

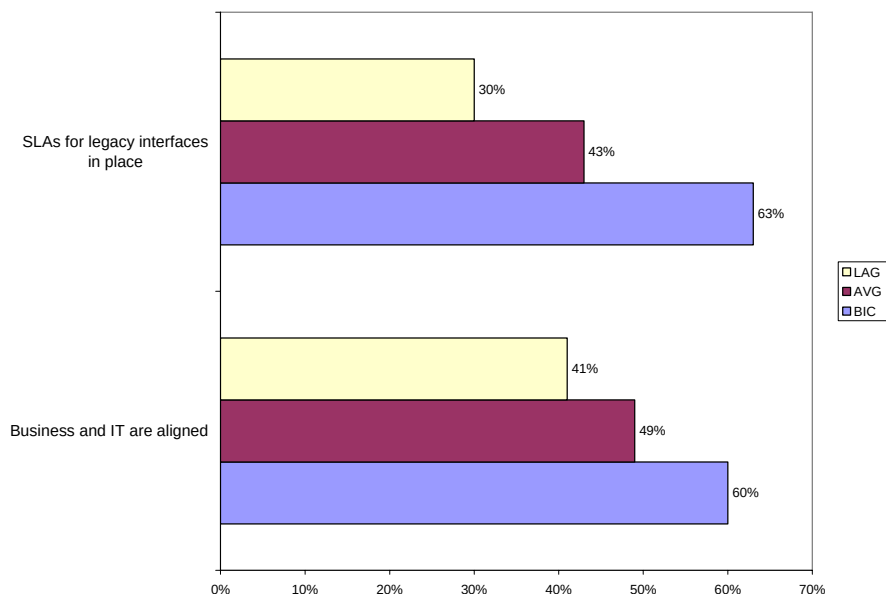


Source: Aberdeen Group, June 2007

Organization

Organizational maturity is reflected in how well IT interfaces with other departments in the company. This is especially important in the area of legacy modernization: IT is touching what is likely a key piece of the business. Service-level agreements (SLAs) are a way to make certain that both IT and the business agree on what will be delivered as part of the modernization effort, and how it will perform. Best In Class companies are using this tool by a margin of **two to one** over Laggards, and 50% more often than the Industry Average (Figure 5).

Figure 6: Agreement among departments drives success



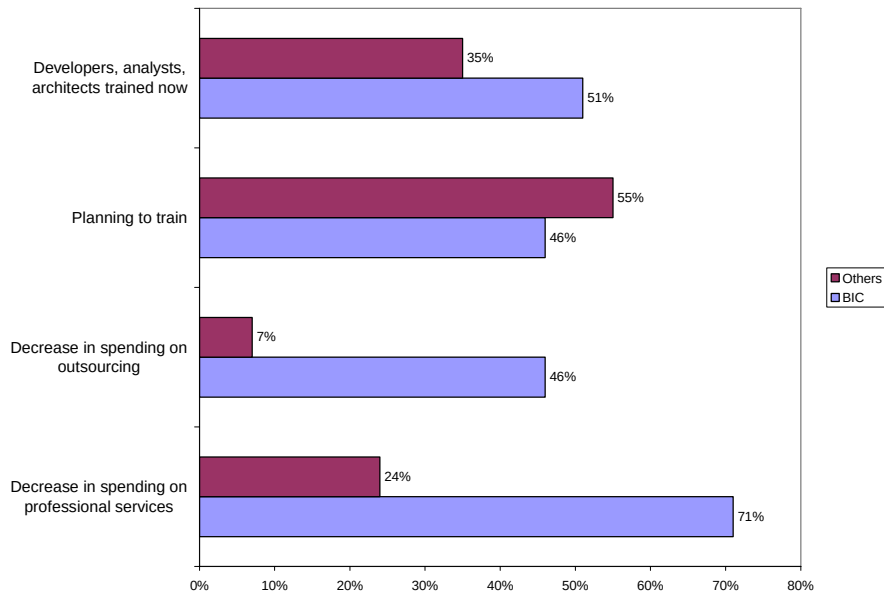
Source: Aberdeen Group, June 2007

Knowledge

Both SOA and web services application development pose unique challenges to the skills of the department, and adding legacy languages and databases to the mix further complicates the issue.

Best In Class organizations are well along in their training efforts, with both Laggard and Industry Average following by six months. Where the knowledge resides is vastly different. Best In Class companies are drastically reducing their spending levels on outsourcing and professional services, while no such reduction was reported for the other two groups (Figure 6).

Figure 7: Training of developers, analysts, and architects



Source: Aberdeen Group, June 2007

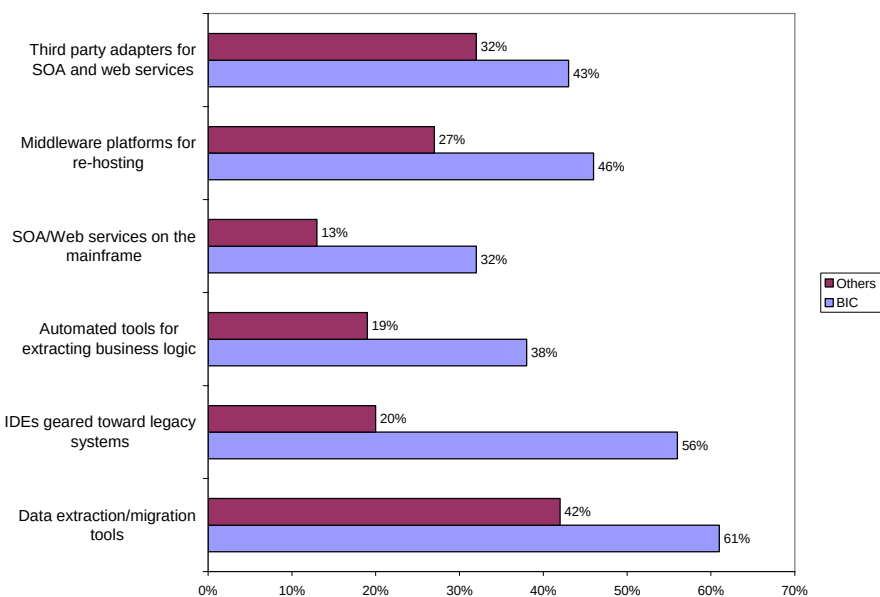
Technology

Best In Class organizations are relying on automation to assist them in extracting both business logic (61%) and data structures (38%, **double the rate** of Laggard and Industry average) as they bring their legacy applications into line with their SOA and web services architecture. This is reflected in the high percentage of Best In Class having documented their legacy interfaces; one of the byproducts of extraction tools is a complete specification of the underlying code or data.

Alongside their modernization efforts, Best In Class are continuing to invest in SOA infrastructure. **Twice as many** Best In Class companies are deploying SOA service layers on the mainframe, with **48%** indicating that spending on SOA platforms is their number one or two funding priority over the next 12 months.

One of the challenges of SOA applications is in testing them. The QA environment is quite a bit different than that for traditional applications. Across the board, all groups indicate that deploying new testing methodologies is a high priority in the next twelve months.

Figure 8: Technical approaches to legacy modernization



Source: Aberdeen Group, June 2007

Aberdeen Insights – Technology

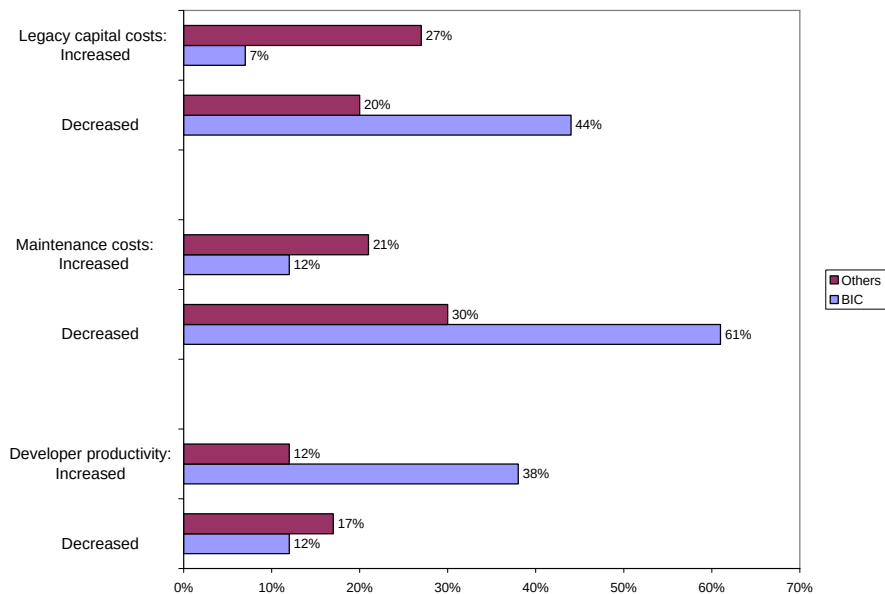
Best In Class companies understand that legacy applications can be especially challenging to bring into line with more modern architectures. They are using automation to speed up the design and development cycle and fully document packages and data structures, and are leveraging third-party adapters for commercial applications.

Legacy specific development environments are also a key to peak performance, with Best In Class organizations using them at double the rate of other groups. These IDEs make it easier for developers to focus on the business logic of the application rather than the mechanics of developing on the legacy platform.

Performance

Overall, Best In Class organizations reported decreases in costs for development, maintenance, and infrastructure (Figure 8). Productivity of mainframe developers was improved by use of specialized IDEs and third-party adapters that expose legacy data and packages. This emphasis on productivity is reflected by the Best In Class' emphasis on in-house application development, 57% versus 28% of the Industry Average.

Figure 9: Best In Class see improvements in costs and productivity



Source: Aberdeen Group, June 2007

Chapter Three: Required Actions

Based on the information in Chapters One and Two, readers should determine whether their organizations fall into the “Laggard”, “Industry Average,” or “Best in Class” categories. The key to gaining value from this report is moving your organization along the maturity path to Best in Class. The following actions will help spur necessary performance improvements:

Laggard Steps to Success

- **Automate.** Use automated extraction tools to pull apart your legacy applications. They will provide clean interfaces to the underlying code, and will let you see what is really going on in each package.
- **Understand what you are delivering.** Best In Class organizations know that service level agreements are not just for B2B relationships. Use SLAs to provide a basis for measuring the performance of individual services. The SLA lets you focus on those components that need the most attention.
- **Use adapters.** Third-party adapters, especially for commercial vertical applications, will save both time and money in your modernization efforts. Best In Class companies use adapters for commodity packages, and focus their efforts on the custom parts of the application.

Industry Norm Steps to Success

- **Document the interfaces.** Spend the time to thoroughly document the packages and data structures of your legacy applications. Whether you opt for rehosting or extending, this step will pay off in
- **Don't focus just on cost.** Getting code off of the mainframe will almost certainly save money in the short term. Best In Class organizations, however, understand that cost is not the only component of a successful architecture. Look to agility as a key driver for your modernization efforts.
- **Switch development tools.** Integrated development environments that are designed for legacy systems will dramatically improve the productivity of your application developers. Forty percent of Best In Class report an increase in productivity, and 56% say that they use legacy IDEs, versus 12% of Industry Average reporting improvement, and 35% using IDEs.

Best in Class Steps to Success

- **Continue to look at governance.** Both run-time and design-time governance help unlock the promise of SOA. Without governance, SOA environments quickly devolve into a morass of functionally similar objects with multiple, possibly incompatible, versions. Change control is critical in a fully evolved SOA environment. Just under half of Best In Class organizations cite governance as their top concern.
- **Re-tool QA and test processes.** Functional testing isn't enough for SOA applications; integration testing is an absolute requirement. Forward-thinking IT organizations are starting to reshape their thinking around QA to encompass the complexity of SOA, and realize that integration testing must be built into the design of new services.
- **Focus on performance.** As SOA deployments become more complex, bottlenecks will start to develop. Scalability and performance of SOA applications are complicated by the black-box approach to SOA components. Investing in performance methodologies now will pay off in improvements in SLAs and throughput later.

Aberdeen Insights – Summary

SOA and web services architectures can be complex, and the addition of legacy application and processes can further complicate the issue. While it may seem overwhelming, Best In Class organizations use process and technology to succeed. Automated extraction tools and defined interfaces let them understand the application landscape and focus on the key components of their legacy packages. SLAs help them manage the services created from the legacy packages.

What is perfectly clear is that mainframes, COBOL, and decades-old code are not going away. The core business processes encapsulated in that 25-year-old code are being given new life in high-performance message-based architectures.

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Appendix A: Research Methodology

Between April and May 2007, Aberdeen Group examined the responses of more than 300 companies across a variety of geographies, industries and company revenues.

Aberdeen supplemented this online survey effort with telephone interviews with select survey respondents, gathering additional information on legacy modernization strategies, experiences, and results.

The study aimed to identify emerging best practices for legacy application modernization projects, and provide a framework by which readers could assess their own organization's capabilities and maturity.

Responding enterprises included the following:

- **Job title/function:** About two-thirds of the survey respondents are in their organizations' IT departments. The research sample included respondents with the following job titles: senior executive (CEO, COO, CFO, VP) (20%); CIO (7%); IT manager or director (42%); internal consultant (15%), and IT staff (13%).
- **Industry:** The research sample included respondents predominantly from high-technology industries and companies. These represented 27% of the sample. A significant number of respondents were in the insurance industries (11%), and finance and banking (17%).
- **Geography:** The survey respondents were distributed: North America (U.S., Canada, Mexico) 62%; Central and South America 2%; Asia/Pacific 16%; Europe, Middle East, and Africa 20%.
- **Company size:** About 25% of respondents were from large enterprises (annual revenues above US\$1 billion, including 14% over US\$5 billion); 35% were from midsize enterprises (annual revenues between \$50 million and \$1 billion); and 40% of respondents were from small businesses (annual revenues of \$50 million or less).

Solution providers recognized as sponsors of this report were solicited after the fact and had no substantive influence on the direction of the report. Their sponsorship has made it possible for Aberdeen Group to make these findings available to readers at no charge.

Table 4: PACE Framework

PACE Key
Aberdeen applies a methodology to benchmark research that evaluates the business pressures, actions, capabilities, and enablers (PACE) that indicate corporate behavior in specific business processes. These terms are defined as follows: Pressures — external forces that impact an organization's market position, competitiveness, or business operations (e.g., economic, political and regulatory, technology, changing customer preferences, competitive) Actions — the strategic approaches that an organization takes in response to industry pressures (e.g., align the corporate business model to leverage industry opportunities, such as product/service strategy, target markets, financial strategy, go-to-market, and sales strategy) Capabilities — the business process competencies required to execute corporate strategy (e.g., skilled people, brand, market positioning, viable products/services, ecosystem partners, financing) Enablers — the key functionality of technology solutions required to support the organization's enabling business practices (e.g., development platform, applications, network connectivity, user interface, training and support, partner interfaces, data cleansing, and management)

Source: Aberdeen Group, June 2007

Table 5: Maturity Framework

Maturity Framework Key
The Aberdeen Maturity Framework defines enterprises as falling into one of the following three levels of practices and performance: Best in class (20%) — Application development practices that are the best currently being employed and significantly superior to the industry norm, and result in the top industry performance. Industry norm (50%) — Application development practices that represent the average or norm, and result in average industry performance. Laggards (30%) — Application development practices that are significantly behind the average of the industry, and result in below average performance In the following categories: Process — What is the scope of process standardization? What is the efficiency and effectiveness of this process? Organization — How is your company currently organized to manage and optimize this particular process? Knowledge — What visibility do you have into key data and intelligence required to manage this process? Technology — What level of automation have you used to support this process? How is this automation integrated and aligned? Performance — What do you measure? How frequently? What's your actual performance?

Source: Aberdeen Group, June 2007

Table 6: Relationship between PACE and Competitive Framework

PACE and Competitive Framework How They Interact

Aberdeen research indicates that companies that identify the most impactful pressures and take the most transformational and effective actions are most likely to achieve superior performance. The level of competitive performance that a company achieves is strongly determined by the PACE choices that they make and how well they execute.

Source: Aberdeen Group, June 2007

Appendix B: Related Aberdeen Research

Related Aberdeen research that forms a companion or reference to this report include:

- Service-Oriented Architectures (October 2005)
- The SOA in IT Benchmark Report (December 2005)
- The Legacy Application Migration Benchmark Report (September 2006)
- The Composite Applications Benchmark Report (December 2006)
- Management and Governance: Planning for an Optimized SOA Application Lifecycle (March 2006)
- SOA Middleware Starts Where Web Services Leave Off (June 2007)

Information on these and any other Aberdeen publications can be found at www.Aberdeen.com.

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